

Yixin Zhang 张艺馨

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English proficiency: IELTS 8.0, CET6 646

Job Experience

R&D Engineer @ 上海县识智能有限公司

Sep 2025 – Now

Startup founded by [Prof. Zhijun Li 李治军](#), and backed by [National Innovation Center par Excellence](#)

- Architected media I/O runtime for a Humanoid Robots OS (InsightOS), supporting egocentric data collection from camera/depth camera sources.
 - Resolved 3 bottlenecks: device discovery, sync between RGB-D depth and color frames, and stream routing/reuse. Eliminated C++ code recompilation between hardware swaps.
 - Engineered agent skills to accelerate AI-driven media app development. Enabled >95% of vision/streaming pipelines to be built within 15 minutes.
- Upgraded a proprietary robot's motion control from open-loop to closed-loop IK by introducing visual and tactile feedback, enabling autonomous electrical panel operation for the WAIC 2026 exhibition.
 - Developed a real-time visual servoing pipeline utilizing YOLO11 segmentation and RGB-D data, dynamically resolving camera-to-robot extrinsics through ROS TF trees.
 - Optimized 6D target object pose estimation from 3D point clouds using SVD, leveraging the known vertical planar constraints of the target buttons.
 - Designed a dexterous hand manipulation runtime, implementing FK/IK across arm and hand joints, to allow for visual servoing.

Industrial Digitalization Engineer @ 北京中科圣泰环境科技有限公司

Jul 2025 – Sep 2025

Established by Ansteel and Institute of Process Engineering, Chinese Academy of Sciences

- Designed a computer vision proof-of-concept for industrial wastewater coagulation dose adjustment, framing the task as a 5-class classification problem (-20% to +20% dose) using labeled on-site video data.
- Achieved 89% offline accuracy with XGBoost using time-series splits to prevent data leakage caused by video frame autocorrelation.

(Internship) Scientist @ SIMTech, A*STAR Singapore

Sep 2024 – Jan 2025

Dynamic Decarbonisation Analytics, advised by [Dr. Aniq Ahsan](#) and [Dr. Aloisius Rabata Purnama](#)

- Worked on the analytics backend behind [Global Mitigation Potential Atlas](#), a climate change mitigation platform developed with public-sector partners from Singapore, the Netherlands, and UN institutions.
- Translated ASEAN energy-system case studies into convex-optimization-ready JSON format for interactive analysis. Visualized and validated the results with prior OSeMOSYS and TIMES studies.
- Prototyped a GenAI workflow to automate json conversion. Reached Pass@1 > 85% with ChatGPT o1.
- Aided in conceptualizing [MOYA Cascade](#), now a launched company/supply-chain level product, featuring emissions data collection and ESG reporting without exposing sensitive supplier data.

Technical Skills

Programming Languages: C++, Python, C#, Matlab

Knowledge about Agentic AI & LLM Workflows:

- Heavy user of GitHub Copilot/SDK and Codex cli; Developing agent skills to automate daily jobs.
- Harness and single/multi-agent architectures including ReAct, Ralph Loop, Agent teams/subagent.
- Agent Communication with A2A/MPP/x402 protocol. Active user of decentralized Agent Network Protocol (ANP) with a [zyx.awiki.ai](#) handle for E2EE agent messaging.

Open source contributor to [CAMEL \(pr#3758\)](#), [FastGPT \(pr#6348\)](#), working with OceanBase, a vector database; and core contributor to [LLM-Cookbook](#), pioneer Chinese tutorial for Prompt Engineering in 2023.

Multiple Certificates in Gen AI, DevOps, Data Science, and Power BI Data Analyst.

Education Background

Institute of Systems Science, National University of Singapore

In person: Jun 2024 – Mar 2025

Graduate Diploma in Systems Analysis | GPA: 4.61/ 5.0

Online: Jun 2023 – Jun 2024

- Coursework: Business requirement analysis, SDLC, Coding with C# and Java for web and mobile apps.
- Shared with the incoming cohort of 40+ students on modern programming, teamwork and keeping pace.

School of Electronics and Information Engineering, Harbin Institute of Technology

B.Eng. in Intelligent Test & Control Engineering (哈工大电信学院智能测控工程) Sep 2020 – Jun 2024

Average Grade: 86.95/ 100 (GPA: 3.38/ 4.0)

- University-wide awards:
 - Recipient of People's Scholarship for 5 semesters, Outstanding Member (优秀团员) (top 3%),
 - Outstanding Individual in Social Practices (top 5%), Star Reader Award (top 10 per year)

Project Experience

Deep Reinforcement Learning-Based Path Planning *Capstone project, HIT*

Sep 2023 – Jun 2024

- Enhanced a Deep Q-Network (DQN) by introducing delays to observation updates and a goal-directed reward mechanism for model-free dynamic path planning, featuring real-time obstacle avoidance.
- Achieved ~12% faster convergence and a 5.9% higher success rate compared to baseline DQN models like Rainbow and D3QN on Gym (simulated environment).

iFlytek SparkDesk LLM Eval: Performance & Security in KBQA

Oct 2023 – Nov 2023

- Crafted a domain-specific agricultural knowledge base to augment the SparkDesk LLM using RAG.
- Strategized team-based adversarial attack and defense, supervised by Prof. Jie Liu 刘劼. Formulated 100+ closed- and open-ended questions mixed with 10+ prompt injection and jailbreak attacks. Assessed model accuracy and analyzed vulnerabilities revealed. [Link to report](#) (in Chinese)

Video Text Extraction System based on CNN and Attention

Sep 2021 – Nov 2022

Team member 1st prize winner and selected national-level, Innovation and Entrepreneurship Programme, HIT

- Architected a video text deblurring and recognition pipeline in 4 stages: text detection, frame tracking, multi-frame deblurring, and final text recognition. Engineered the deblurring module with inspiration from BasicSR
- Optimized the system for lightweight deployment on old Android devices, for scenarios like capturing fast-moving license plates and street views, plus extracting on-screen subtitles from short videos.
- Coordinated meetings and executed project management with Gantt chart, ensuring 90% adherence to planned roadmap.

Social Practice and Volunteering

Beach Cleanup with SG Beach Warriors

Jun 2024 – Jan 2025

Local Non-Profit Organization in Singapore, spotlighted by Lianhe Zaobao

- ▶ Collaborated in coastal conservation efforts across 2 major shorelines (North and East beach), jointly cleaned over 5 tons of marine trash at North Coast as a team of 28 and 63.
- “Exploring the Pillars of a Great Power” *Team leader* Jun 2022 – Sep 2022
 - 2nd prize winner and selected key incubation team, HIT
 - ▶ Conducted field research at Space Museum, World 5G Convention 2022 in a group of 9, and organized individual research at local industrial enterprises and regional museums in team members' hometown.
 - ▶ Compiled an report on the regional equipment manufacturing based on compiled data and interviews.
- Contributed to official bilibili channel of Veritasium, which makes STEM concepts accessible to the public. Translation work include video.
- Volunteered for for major institutional events in HIT, including the university graduation ceremony and milestone decennial alumni reunions.